

"Analysis of Student Academic Performance "

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R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com> (<http://rmarkdown.rstudio.com>).

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
student_data <- read.csv("student_performance.csv")
student_data
```

##	Student_ID	Gender	Attendance	Study_Hours	Assignment_Score	Exam_Marks	Result
## 1	101	Male	85	3	78	82	Pass
## 2	102	Female	92	5	88	91	Pass
## 3	103	Male	65	2	60	55	Pass
## 4	104	Female	88	4	82	85	Pass
## 5	105	Male	72	3	70	68	Pass
## 6	106	Female	95	6	92	94	Pass
## 7	107	Male	58	1	52	45	Pass
## 8	108	Female	81	4	75	79	Pass
## 9	109	Male	76	3	68	72	Pass
## 10	110	Female	90	5	85	89	Pass
## 11	111	Male	68	2	62	58	Pass
## 12	112	Female	87	4	80	83	Pass
## 13	113	Male	74	3	72	70	Pass
## 14	114	Female	93	6	90	92	Pass
## 15	115	Male	61	2	55	52	Pass
## 16	116	Female	84	4	78	80	Pass
## 17	117	Male	79	3	74	76	Pass
## 18	118	Female	89	5	86	87	Pass
## 19	119	Male	70	2	65	63	Pass
## 20	120	Female	96	6	94	96	Pass
## 21	121	Male	83	4	81	84	Pass
## 22	122	Female	77	3	73	74	Pass
## 23	123	Male	91	5	89	90	Pass
## 24	124	Female	69	2	59	57	Pass
## 25	125	Male	86	4	83	86	Pass
## 26	126	Female	94	5	91	93	Pass
## 27	127	Male	63	1	50	48	Pass
## 28	128	Female	80	3	71	73	Pass
## 29	129	Male	75	3	67	69	Pass
## 30	130	Female	88	4	79	81	Pass

```
head(student_data)
```

##	Student_ID	Gender	Attendance	Study_Hours	Assignment_Score	Exam_Marks	Result
## 1	101	Male	85	3	78	82	Pass
## 2	102	Female	92	5	88	91	Pass
## 3	103	Male	65	2	60	55	Pass
## 4	104	Female	88	4	82	85	Pass
## 5	105	Male	72	3	70	68	Pass
## 6	106	Female	95	6	92	94	Pass

```
summary(student_data)
```

```
##      Student_ID      Gender      Attendance      Study_Hours
## Min.   :101.0    Length   :30    Min.   :58.00    Min.   :1.000
## 1st Qu.:108.2    N.unique  : 2    1st Qu.:72.50    1st Qu.:3.000
## Median :115.5    N.blank   : 0    Median :82.00    Median :3.500
## Mean   :115.5    Min.nchar: 4    Mean   :80.30    Mean   :3.567
## 3rd Qu.:122.8    Max.nchar: 6    3rd Qu.:88.75    3rd Qu.:4.750
## Max.   :130.0                                Max.   :96.00    Max.   :6.000
## Assignment_Score Exam_Marks      Result
## Min.   :50.00    Min.   :45.00    Length   :30
## 1st Qu.:67.25    1st Qu.:68.25    N.unique  : 1
## Median :76.50    Median :79.50    N.blank   : 0
## Mean   :74.97    Mean   :75.73    Min.nchar: 4
## 3rd Qu.:84.50    3rd Qu.:86.75    Max.nchar: 4
## Max.   :94.00    Max.   :96.00
```

```
str(student_data)
```

```
## 'data.frame':   30 obs. of  7 variables:
## $ Student_ID      : int  101 102 103 104 105 106 107 108 109 110 ...
## $ Gender          : chr  "Male" "Female" "Male" "Female" ...
## $ Attendance      : int  85 92 65 88 72 95 58 81 76 90 ...
## $ Study_Hours     : int   3 5 2 4 3 6 1 4 3 5 ...
## $ Assignment_Score: int   78 88 60 82 70 92 52 75 68 85 ...
## $ Exam_Marks      : int   82 91 55 85 68 94 45 79 72 89 ...
## $ Result          : chr  "Pass" "Pass" "Pass" "Pass" ...
```

```
colSums(is.na(student_data))
```

```
##      Student_ID      Gender      Attendance      Study_Hours
##           0           0           0           0
## Assignment_Score Exam_Marks      Result
##           0           0           0
```

```
sum(duplicated(student_data))
```

```
## [1] 0
```

```
median(student_data$Exam_Marks)
```

```
## [1] 79.5
```

```
mean(student_data$Exam_Marks)
```

```
## [1] 75.73333
```

```
min(student_data$Exam_Marks)
```

```
## [1] 45
```

```
max(student_data$Exam_Marks)
```

```
## [1] 96
```

```
summary(student_data$Exam_Marks)
```

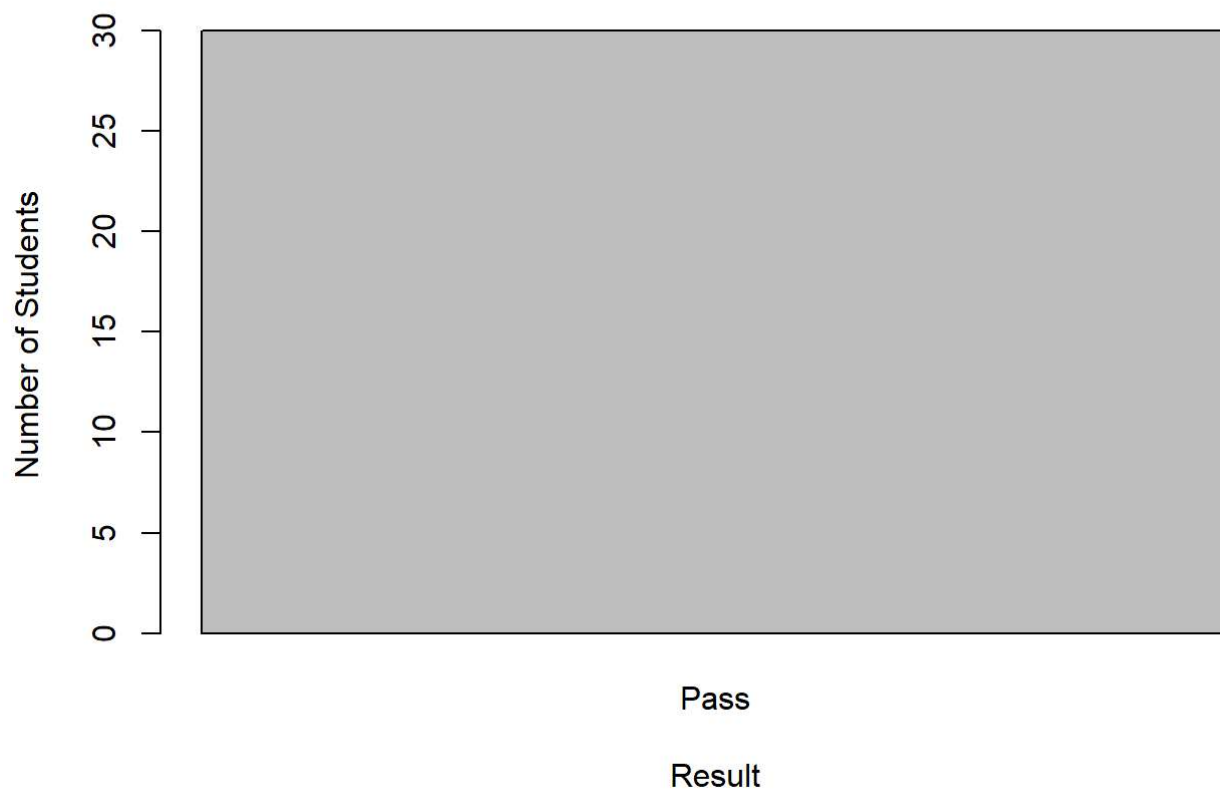
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  45.00   68.25   79.50   75.73   86.75   96.00
```

```
student_count <- table(student_data$Result)
student_count
```

```
##
## Pass
##    30
```

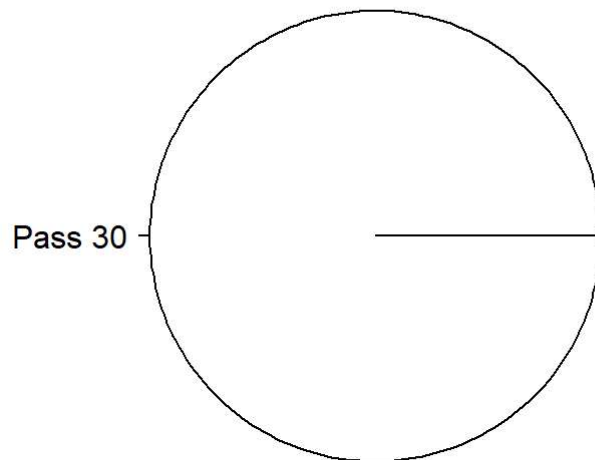
```
barplot(student_count,
        main= "Pass and Fail Student",
        xlab="Result",
        ylab="Number of Students")
```

Pass and Fail Student



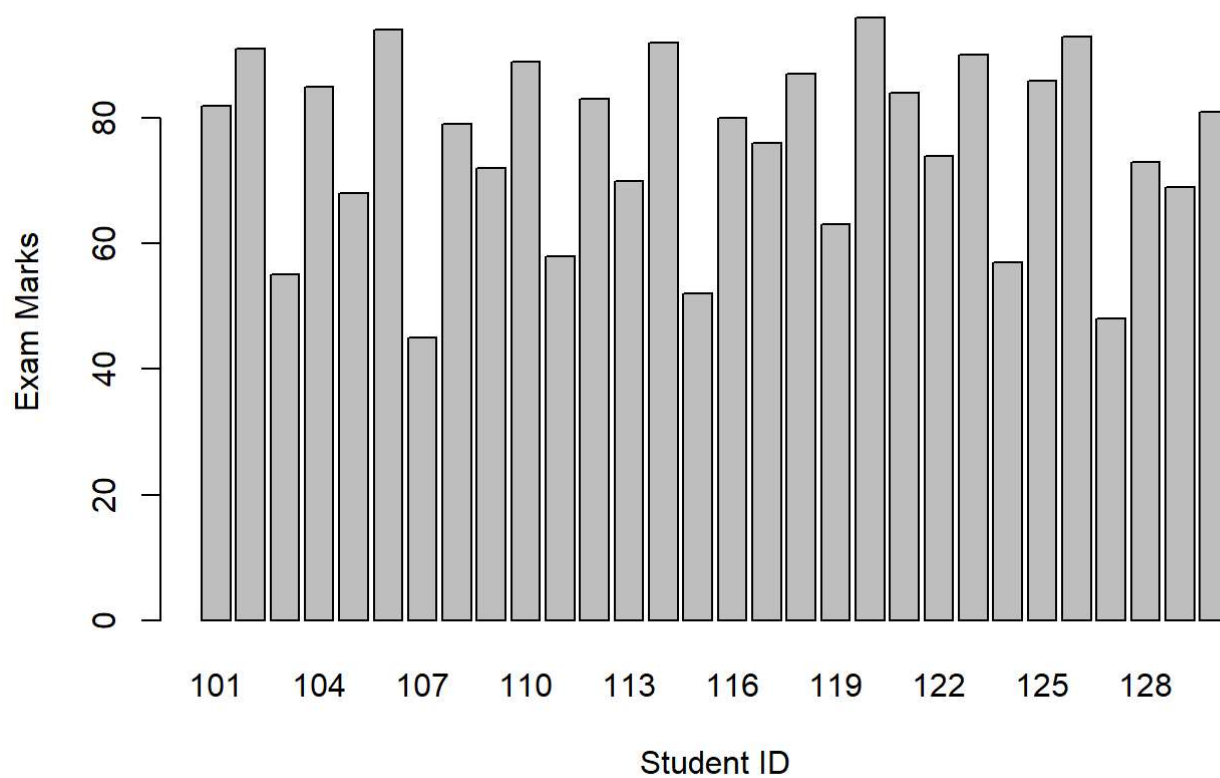
```
pie(student_count,  
    main = "Student Pass and Fail  
Distribution",  
    labels = paste(names(student_count),  
                    student_count))
```

Student Pass and Fail Distribution

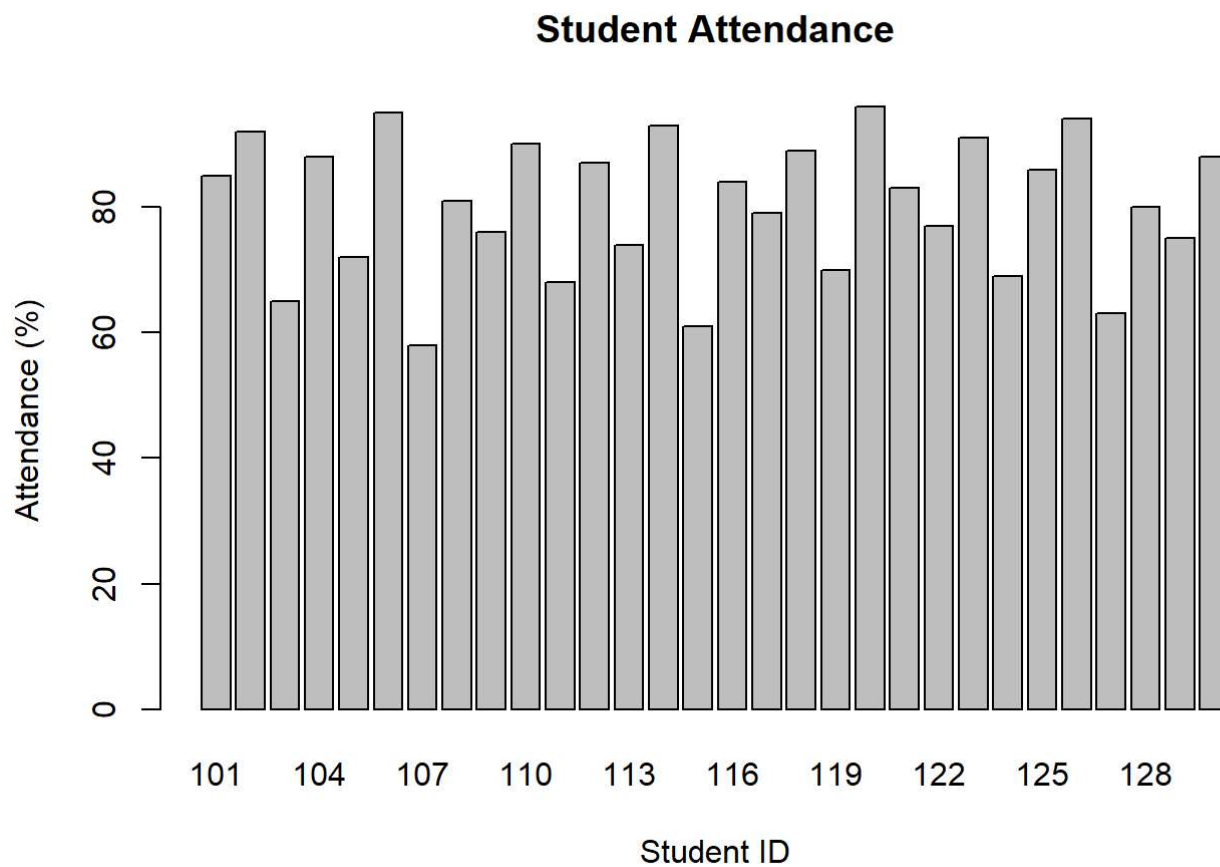


```
barplot(student_data$Exam_Marks,  
        names.arg = student_data$Student_ID,  
        main = "Exam Marks of Students",  
        xlab = "Student ID",  
        ylab = "Exam Marks")
```

Exam Marks of Students



```
barplot(student_data$Attendance,  
        names.arg = student_data$Student_ID,  
        main = "Student Attendance",  
        xlab = "Student ID",  
        ylab = "Attendance (%)")
```

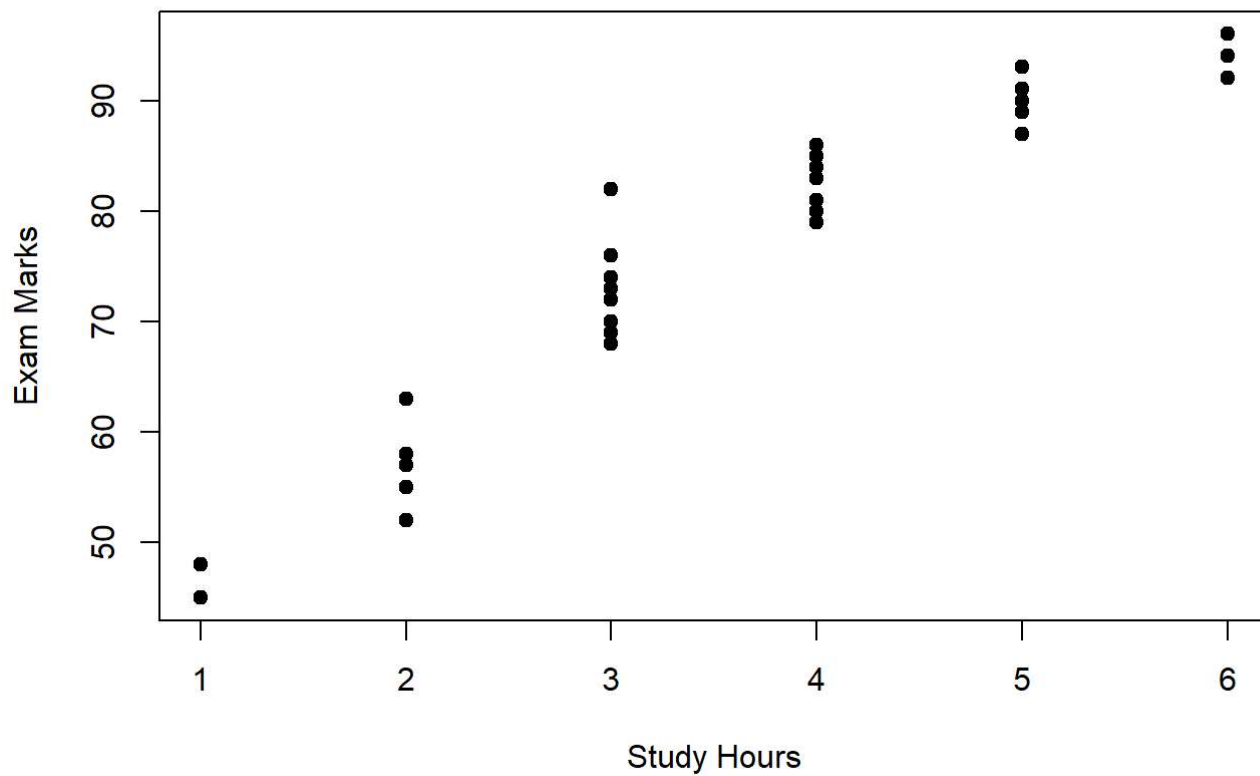


Including Plots

You can also embed plots, for example:

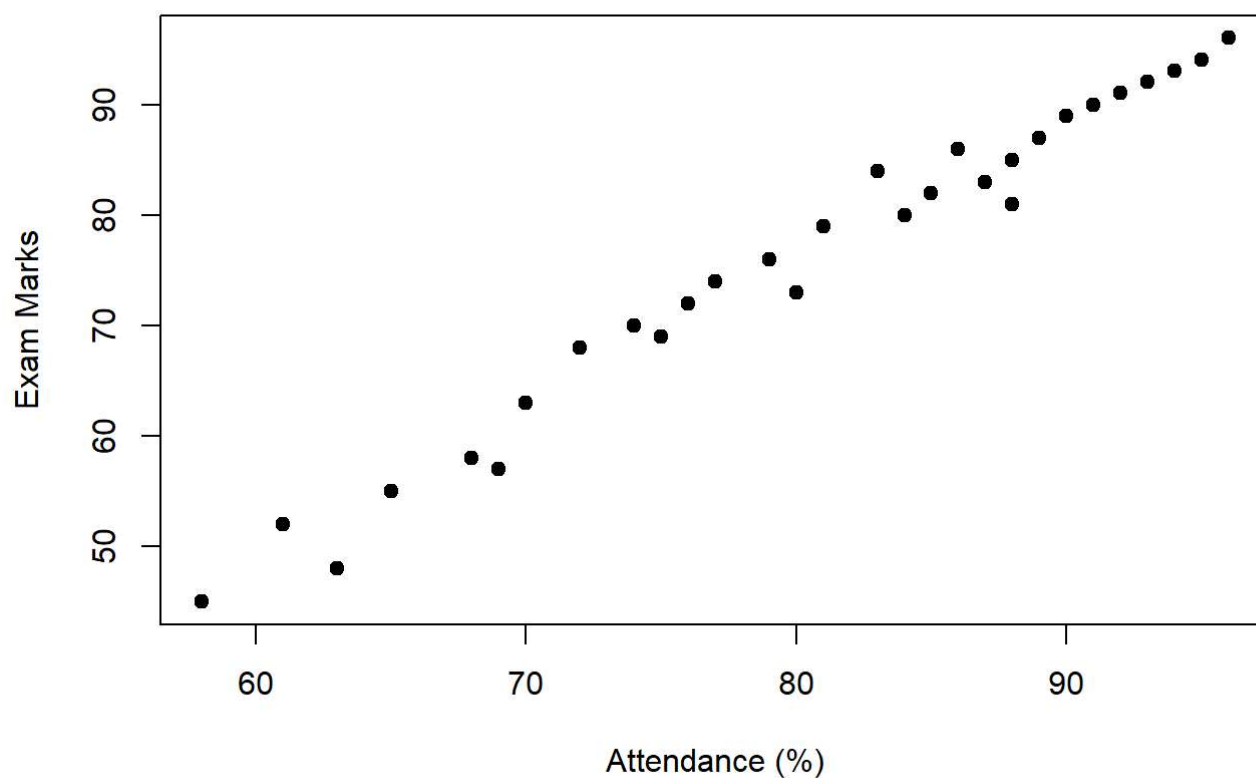
```
plot(student_data$Study_Hours,  
      student_data$Exam_Marks,  
      main = "Study Hours vs Exam Marks",  
      xlab = "Study Hours",  
      ylab = "Exam Marks",  
      pch = 19)
```

Study Hours vs Exam Marks



```
plot(student_data$Attendance,  
      student_data$Exam_Marks,  
      main = "Attendance vs Exam Marks",  
      xlab = "Attendance (%)",  
      ylab = "Exam Marks",  
      pch = 19)
```

Attendance vs Exam Marks



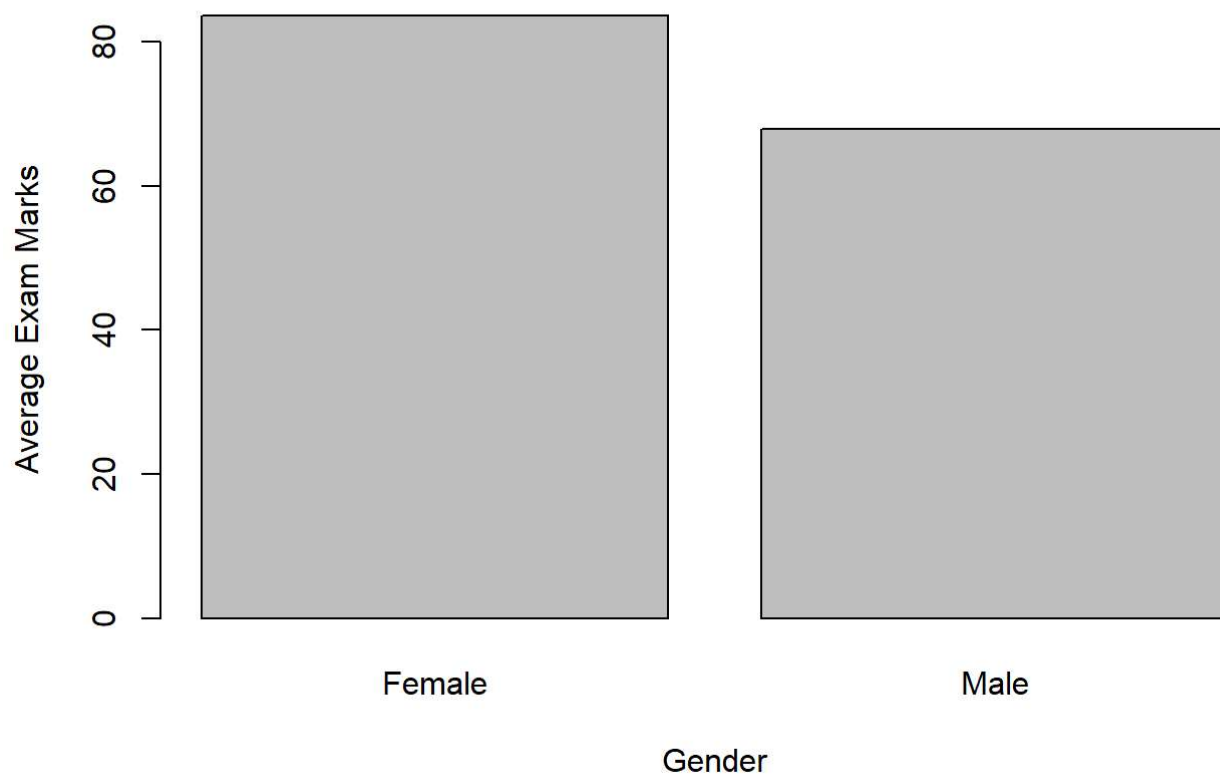
```
aggregate(Exam_Marks ~ Gender,
          data = student_data,
          FUN = mean)
```

```
##   Gender Exam_Marks
## 1 Female   83.60000
## 2  Male    67.86667
```

```
gender_average <- aggregate(Exam_Marks ~ Gender,
                             data =student_data,
                             FUN = mean)
```

```
barplot(gender_average$Exam_Marks,
        names.arg = gender_average$Gender,
        main = "Average Exam Marks by
Gender",
        xlab = "Gender",
        ylab = "Average Exam Marks")
```

Average Exam Marks by Gender



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.